What Can Repeated Size Profiles Identify? Bayesian Inference and Confounding in Inverse Integral Projection Models

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June 17, 2026

Abstract

Repeated cross-sectional size profiles—counts of individuals by body size collected over several years—are among the most common and least expensive data in population ecology. It is tempting to invert a size-structured integral projection model (IPM) against such data and read off survival, growth, and reproduction. We ask instead a prior question: *which* vital rates are identified by repeated profiles, which are confounded, and what additional information resolves them. Using an individual-based generator distinct from the recovery model, a positive numerical-quadrature inverse IPM, and a fully Bayesian treatment, we show that size profiles reliably recover survival, the growth increment, and the recruit-size distribution, but that recruitment intensity is only weakly identified, along a single ridge in parameter space. We prove two identifiability results—a decomposition result (the total per-capita contribution of each size is identified but its split into survival and recruitment is not, wherever growth and recruit sizes overlap) and an excitation result (near a stable size distribution, profiles constrain the asymptotic growth rate λ and the stable shape far more tightly than the vital rates that generate them)—and confirm both numerically. We calibrate the posterior approximation by simulation-based calibration, quantify how size-dependent detection corrupts recovery, and measure how much auxiliary individual-level data are required to break the confounding. A 20-year brook trout capture–mark–recapture study illustrates the central caution: a profile-only fit can look entirely plausible yet badly misestimate growth and the size pattern of survival, which only the individual data reveal. We frame the contribution as a diagnostic framework—what is identified, how to detect confounding, and how to design the data or priors needed for credible inference—rather than a claim that profiles generally identify vital rates.

Keywords: integral projection models; identifiability; inverse problems; Bayesian hierarchical models; integrated population models; simulation-based calibration; size-structured populations.

1 Introduction

Size-structured populations are routinely monitored by repeated cross-sectional surveys that record the body sizes of sampled individuals. The resulting *size profiles*—histograms of size at successive census times—are inexpensive relative to individual marking, and they exist for many taxa over

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long periods. The integral projection model (Easterling et al., 2000; Ellner and Rees, 2006; Merow et al., 2014) provides a natural mechanistic description of how such a profile should evolve: a kernel built from size-dependent survival, growth, and reproduction maps this year's size distribution to next year's. A seemingly direct strategy is therefore to invert the IPM against repeated profiles and recover the vital-rate functions. This is an instance of a classic inverse problem for structured populations (Wood, 1994; Gross et al., 2002; Ghosh et al., 2012), and like most inverse problems it is ill-posed: many different vital-rate configurations can generate nearly identical profile dynamics.

The central message of this paper is that the correct scientific question is not "can we fit an inverse IPM to size profiles" (we almost always can) but "*what does the fit identify.*" A model can reproduce observed profile dynamics accurately while estimating a population-effective transition operator that differs from the individual vital rates that generated the data, and it can report narrow posterior intervals that are an artifact of the prior rather than the data. We therefore treat identifiability, confounding, and prior sensitivity as first-class objects of study, declare the estimands before fitting, and validate recovery against data generated by a process *other* than the recovery model.

Our contributions are: (i) a clean separation of an individual-based biological generator from the recovery model, so that the inverse IPM is never validated only on data from the same IPM; (ii) a decomposition of which vital rates are identified by repeated profiles and a characterization—analytical and numerical—of the confounding ridge between recruitment intensity and its size gradient; (iii) a calibrated Bayesian workflow, including simulation-based calibration of the posterior approximation and an explicit demonstration that size-dependent detection, if unmodeled, is mistaken for demography; (iv) a quantitative account of how much auxiliary individual data break the confounding, including robust prior borrowing and its transportability limits; and (v) a real-data application to a long brook trout time series in which profile-only inference is benchmarked against capture–mark–recapture and integrated analyses. The methodological emphasis and the identifiability diagnostics are intended to be portable to other size- or stage-structured inverse problems.

▶ **[COAUTHOR INPUT NEEDED – biology/ecology]:** *One to two paragraphs of ecological and management motivation: why size profiles are collected in your systems, what management or conservation decisions rest on vital-rate estimates from them, and why distinguishing identified from confounded rates matters in practice. Cite the key applied literature for the focal taxa.*

The remainder of the paper declares the estimands and notation (Section 2), specifies the generator, observation process, recovery model, priors, inference, and diagnostics (Section 3), reports the simulation experiments (Section 5), presents the brook trout application (Section 6), and discusses scope and limitations (Section 7).

2 Estimands and Notation

We distinguish four inferential targets and report them separately throughout, because good performance on one does not imply recovery of another:

1. **Individual vital rates:** survival probability $s(x)$, mean and spread of the growth transition $g(x' \mid x)$, realized recruitment intensity $r(x)$, and the recruit-size distribution $b(x' \mid x)$, as functions of individual size x.
2. **Population-effective rates:** the same quantities after averaging over unobserved heterogeneity, movement, and detection—what a size-only model can at best target.
3. **Kernel functionals:** the dominant eigenvalue λ (asymptotic population growth rate), the

stable size distribution, and sensitivities.

4. **Prediction targets:** next-year and multi-year size profiles.

Throughout, x denotes size on a fixed mesh, Δx the mesh spacing, $\boldsymbol{\mu}_t$ the latent population size-intensity at census t, and Y_{tj} the observed count in size bin j at census t.

3 Methods

3.1 Individual-based generator

The data generator operates on individuals, not on the recovery kernel, so that recovery is never tested only against self-generated data. At census t, individual i has size x_{it}, age, sex q_i, and optional persistent latent quality u_i. Survival is an individual Bernoulli event,

$$S_{it} \sim \text{Bernoulli}\{s(x_{it}, q_i, u_i, E_t, N_t)\}, \tag{1}$$

with E_t an environmental covariate and N_t population size. Conditional on survival, next size is drawn from a growth distribution,

$$x_{i,t+1} \mid S_{it} = 1 \sim \mathcal{N}\{m_g(x_{it}), \sigma_g^2\}, \tag{2}$$

and realized recruitment is generated separately as an individual count process with recruit sizes drawn independently of parent size up to its mean,

$$R_{it} \sim \text{Poisson}\{r(x_{it}) \nVdash (q_i = \text{F}) \cdot c\}, \qquad x_{\text{recruit}} \sim \mathcal{N}\{m_b(x_{it}), \sigma_b^2\}. \tag{3}$$

Female reproductive output is scaled so that its population average matches the declared recruitment curve. Survival, growth, and reproduction are thus distinct stochastic events: the simulated population does not evolve by deterministic application of the recovery IPM.

▶ **[COAUTHOR INPUT NEEDED – biology/ecology]:** *Confirm that this individual-level generator matches the biology of the focal systems (e.g. semelparity vs. iteroparity, sexual size dimorphism, density dependence, environmental drivers). Note any structurally important process the generator omits.*

3.2 Size-profile observation process

Identities are retained by the generator but discarded before profile-only analysis. At each census, individuals are independently detected with probability $q_t(x)$, optionally with measurement error, and grouped into equal-width bins. The general observation intensity is $\lambda_t(x) = e_t\, q_t(x)\, \mu_t(x)$ with effort e_t. Conditional on the initial profile, subsequent bin counts follow either a Poisson model $Y_{tj} \sim \text{Poisson}(M_{tj})$ or a Gamma–Poisson model with negative-binomial margin (mean M_{tj}, precision ϕ) that accommodates overdispersion, where M_{tj} is the model-implied expected count.

3.3 Positive inverse IPM

The recovery model propagates latent size intensity through a decomposed, strictly positive kernel,

$$\mu_{t+1}(x') = \int K(x', x)\, \mu_t(x)\, dx, \qquad K(x', x) = P(x', x) + F(x', x), \tag{4}$$

$$P(x', x) = s(x)\, g(x' \mid x), \qquad F(x', x) = r(x)\, b(x' \mid x), \tag{5}$$

with both conditional size densities normalized, $\int g(x' \mid x)\, dx' = \int b(x' \mid x)\, dx' = 1$. Consequently the column integrals of P and F are exactly $s(x)$ and $r(x)$, removing arbitrary scaling between recruitment intensity and recruit-size shape. This normalization does *not* by itself identify the split of K into P and F; that split must be learned from temporal information, restrictions, priors, or auxiliary data. The vital-rate functions are parameterized at a centering size of 80 with $z(x) = (x - 80)/20$:

$$\operatorname{logit} s(x) = \alpha_s + \beta_s z, \qquad m_g(x) - x = \max\{0, \alpha_g + \beta_g z\}, \qquad \sigma_g = e^{\eta_g}, \tag{6}$$

$$\log r(x) = \alpha_r + \beta_r z, \qquad m_b(x) = \alpha_b + \beta_b z, \qquad \sigma_b = e^{\eta_b}, \tag{7}$$

giving ten demographic parameters; the Gamma–Poisson model adds $\log \phi$. The continuous operator is discretized on a regular mesh, $\boldsymbol{\mu}_{t+1} = \Delta x\, \mathbf{K} \boldsymbol{\mu}_t$, with conditional normal transition probabilities *integrated over target bins* rather than evaluated at bin centers (which otherwise inflates recruit-size variance at coarse bins), and renormalized over the retained mesh.

3.4 Priors

We use independent weakly informative normal priors on the transformed coefficients; representative induced 95% prior intervals are biologically diffuse but retain structure through the link functions, parametric curve forms, normalized transition distributions, and the non-negative growth constraint. The Gamma–Poisson precision receives $\log \phi \sim \mathcal{N}\{\log 50, 2^2\}$. External information, when used, is expressed as an explicit auxiliary study design rather than only as abstract prior standard deviations (Section 5.6).

3.5 Inference

Posterior computation uses maximum *a posteriori* optimization with Gaussian Laplace approximations for the broad grids—fast and resumable—audited against adaptive-Metropolis MCMC on representative datasets, and validated by simulation-based calibration (Cook et al., 2006; Talts et al., 2018). Non-positive-definite posterior Hessians are recorded as approximation failures and excluded from coverage summaries. For the finite-population state-space variant we use a particle marginal Metropolis–Hastings sampler (reported in supplementary material); the present paper's claims rest on the deterministic projection likelihood and its audited approximation.

3.6 Identifiability diagnostics

We report, for every fit: (i) prior-to-posterior contraction $1 - \mathrm{sd}_{\mathrm{post}}/\mathrm{sd}_{\mathrm{prior}}$ per parameter, which separates data-informed from prior-dominated quantities; (ii) the posterior correlation between recruitment level α_r and slope β_r, the leading confounded pair; and (iii) the condition number and least-informed eigenvector of the observed information, which expose weakly identified directions even when no single parameter is flat.

4 Identifiability Analysis

Two results explain, and predict, the empirical recovery patterns reported below. Both concern the discretized one-step map $\mathbb{E}[\boldsymbol{\mu}_{t+1}] = \Delta x\, \mathbf{K} \boldsymbol{\mu}_t$ with per-capita kernel entries $K_{ij} = s_j\, g_{ij} + r_j\, b_{ij}$, where column j collects the contributions of source size j to all destination sizes i.

4.1 Decomposition identifiability

Proposition 1 (Total contribution is identified; its split is not, on the overlap)**.** *The column sum $\Delta x \sum_i K_{ij} = s_j + r_j$ equals the total per-capita contribution of source size j, because g and b are normalized. Hence, wherever the kernel column $K_{\cdot j}$ is identified, the sum $s_j + r_j$ is identified. The decomposition into survival-growth $s_j g_{\cdot j}$ and recruitment $r_j b_{\cdot j}$ is identified on destination sizes where the growth density $g(\cdot \mid j)$ and the recruit-size density $b(\cdot \mid j)$ have disjoint support, and is confounded on destination sizes where they overlap.*

Argument. On a destination size i in $\operatorname{supp}(g(\cdot \mid j)) \setminus \operatorname{supp}(b(\cdot \mid j))$ the entry $K_{ij} = s_j g_{ij}$ involves only survival-growth, and symmetrically for recruitment; on the overlap $K_{ij} = s_j g_{ij} + r_j b_{ij}$ constrains only the sum of two positive terms, so (s_j, r_j) and the local density values trade off freely subject to nonnegativity and the normalization constraints. □

This is exactly the pattern we observe. Recruit sizes are small and overlap the sizes that small survivors grow into, so recruitment *intensity* is confounded with small-fish survival and growth; survival of larger sizes, whose grown distribution avoids the recruit band, is identified, as is the recruit-size distribution itself. The leading practical shadow is the recruitment level/slope ridge of Section 5.1 (Figure 1): with all other parameters fixed, (α_r, β_r) trade off along a near-flat direction with posterior correlation below -0.95.

4.2 Excitation and the role of stationarity

Proposition 2 (Near stationarity, only λ and the stable shape are identified)**.** *Suppose the latent profiles are proportional to a fixed shape, $\boldsymbol{\mu}_t = c_t \mathbf{w}$ for all t. Then the expected data depend on the kernel only through $\Delta x \, \mathbf{K}\mathbf{w} = \lambda \mathbf{w}$, with $\lambda = c_{t+1}/c_t$. Any kernel $\mathbf{K}'$ sharing this dominant eigenpair produces identical expected profiles, so the dominant eigenvalue λ and the stable shape $\mathbf{w}$ are identified, but the remaining $m(m-1)$ degrees of freedom of $\mathbf{K}$—and hence the vital rates beyond $(\lambda, \mathbf{w})$—are not. Identifying more of $\mathbf{K}$ requires the population to occupy multiple linearly independent states (excitation: cohort pulses, environmental variation, or transient non-stationary dynamics).*

This makes "a population observed near its stable size distribution carries little information about its generating operator" precise, and predicts that the kernel *functional* λ is recovered far better than the rates that produce it. Section 6.3 confirms this directly in the brook trout data: the profile-only posterior spreads across biologically very different vital-rate regimes that nonetheless share nearly the same λ.

5 Simulation Experiments

5.1 The recruitment confounding ridge

A large, low-noise data set generated from the recovery model itself (one long trajectory, full detection, ∼6000 individuals) isolates *structural* identifiability from Monte Carlo noise. The profile likelihood is extremely peaked in most directions but nearly flat along a single ridge that trades recruitment level against recruitment slope (Figure 1): the posterior correlation between α_r and β_r is -0.96, the curvature of the two-dimensional log-likelihood implies a ridge correlation of 0.98, and the observed-information condition number exceeds 7×10^3. Intuitively, the profile data constrain the total kernel K and the influx into each size bin, but not the partition of that influx between survival-growth flux and recruitment flux—so a higher recruitment level with a flatter size gradient produces nearly the same profile dynamics as a lower level with a steeper gradient.

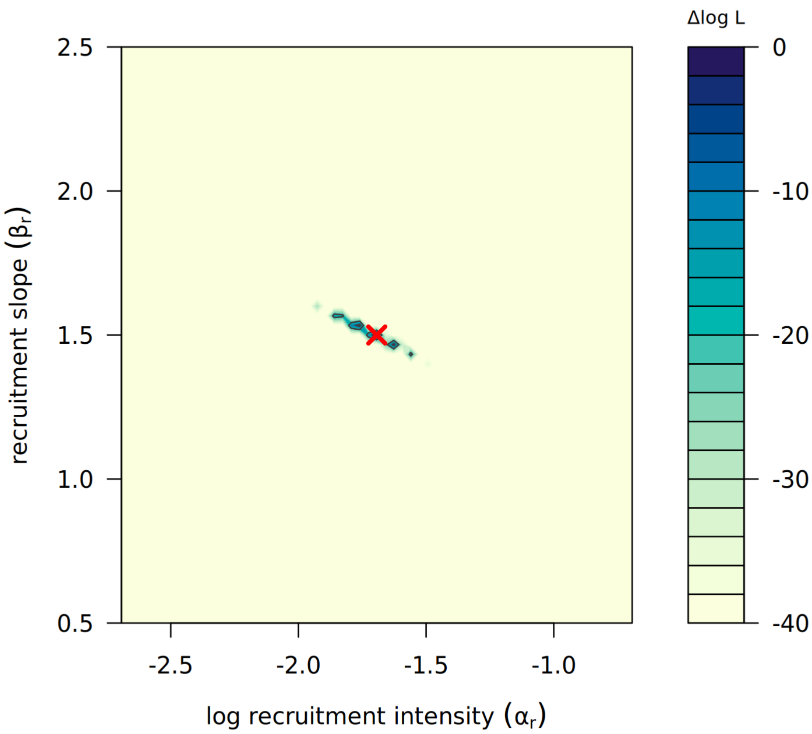

Figure 1: Relative profile log-likelihood over recruitment level α_r and slope β_r, all other parameters at truth (red cross), capped at $\Delta \log L = -40$ to reveal structure. The likelihood is sharply peaked except along a thin ridge oriented in the negative-correlation direction: the non-identified recruitment direction.

5.2 Excitation restores recruitment identifiability

Proposition 2 predicts that the confounding stems from near-stationarity and should weaken as the population is perturbed off its stable distribution. We tested this with matched data sets differing only in the initial state, interpolated from exactly the stable distribution $\mathbf{w}$ to a strong small-size cohort pulse, holding the vital rates, population size, and observation noise fixed (Figure 2). Three patterns emerge. First, the dominant eigenvalue λ is recovered essentially perfectly at *every* excitation level (relative error < 0.01), including at exact stationarity—the asymptotic functional is identified even when the rates are not, the positive half of Proposition 2. Second, survival remains identified throughout (contraction ≈ 0.98). Third, recruitment intensity—nearly unrecoverable near stationarity (relative error ≈ 0.8–1.1, level/slope posterior correlation below -0.99)—is progressively recovered as excitation increases, reaching relative error 0.03 and level/slope correlation -0.31 under the strongest cohort pulse. The confounding is thus *excitation-limited*: a perturbed population (following a disturbance, a strong cohort, or environmental forcing) carries the temporal information needed to separate recruitment from survival-growth, whereas one observed near equilibrium does not. This converts the negative identifiability result into concrete design guidance—to recover recruitment from size profiles, observe the population away from its stable distribution.

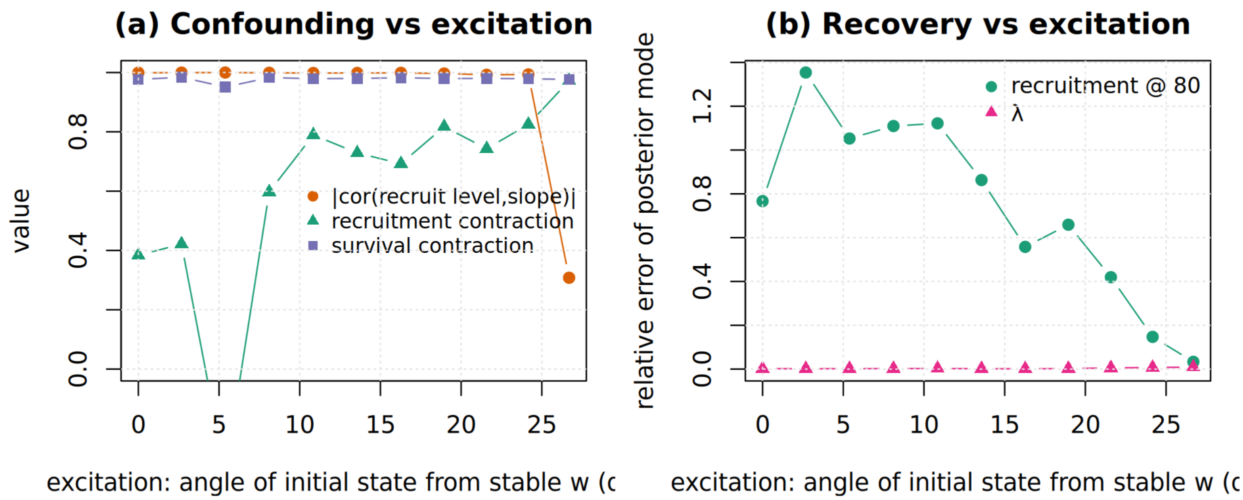

Figure 2: Excitation experiment. As the initial state is perturbed further from the stable distribution (x-axis: angle from $\mathbf{w}$): (a) the recruitment level/slope confounding breaks and recruitment contraction rises, while survival stays identified; (b) recruitment intensity recovery improves by an order of magnitude, while λ is recovered accurately at all excitation levels.

5.3 Broad recovery grid

A factorial Monte Carlo grid varied the allocation of a fixed 12-transition budget (1×12, 3×4, 6×2), initial population size, profile detection, initial profile shape, and recovery likelihood (Poisson vs. Gamma–Poisson), producing 240 simulated datasets and 480 fits. Replicated short trajectories outperformed one long trajectory on average (Table 1), but no design achieved nominal coverage for all quantities. Crucially, recovery was highly quantity-dependent (Figure 3): the recruit-size distribution, survival, and mean growth were recovered well, whereas recruitment intensity and the dominant eigenvalue were biased and overconfident—the simulation counterpart of the ridge in Section 5.1.

Table 1: Mean 95% interval coverage by allocation of the 12-transition budget.

Design	Poisson params	Poisson derived	GP params	GP derived
1×12	0.734	0.701	0.764	0.718
3×4	0.795	0.775	0.818	0.818
6×2	0.795	0.813	0.816	0.836

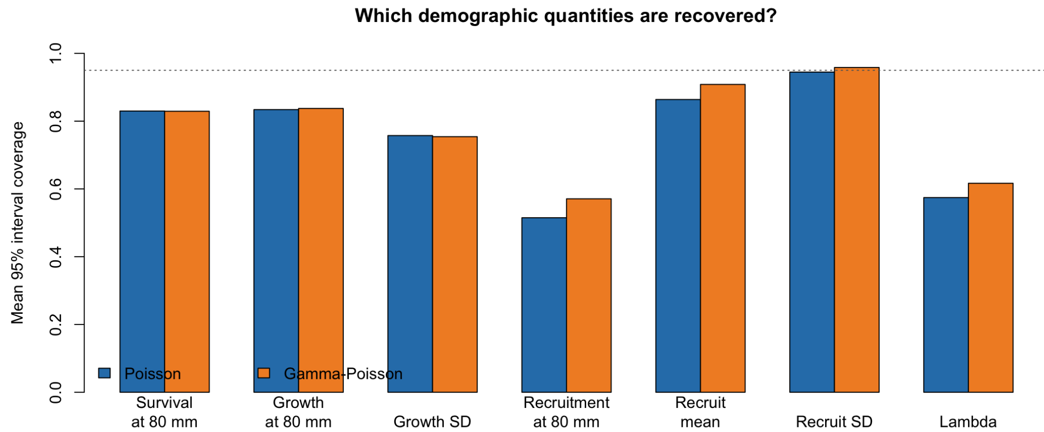

Figure 3: Interval coverage varied sharply among demographic quantities. Recruit size, survival, and mean growth were recovered more reliably than recruitment intensity and the dominant eigenvalue.

5.4 Calibration of the posterior approximation

We validated the Laplace posterior by simulation-based calibration, generating data from the recovery model's own data-generating process so that rank non-uniformity isolates computational and approximation error rather than misspecification (Figure 4). Under an informative prior, 200 of 200 fits succeeded and 10 of 11 parameters had uniform rank statistics (only the overdispersion parameter $\log \phi$ was mildly miscalibrated). Under a diffuse prior the rank histograms became U-shaped—posterior intervals overconfident—most severely along exactly the confounded recruitment and growth directions. The practical implication is that, under diffuse priors, interval inference for the confounded directions should come from full posterior sampling rather than the Gaussian approximation.

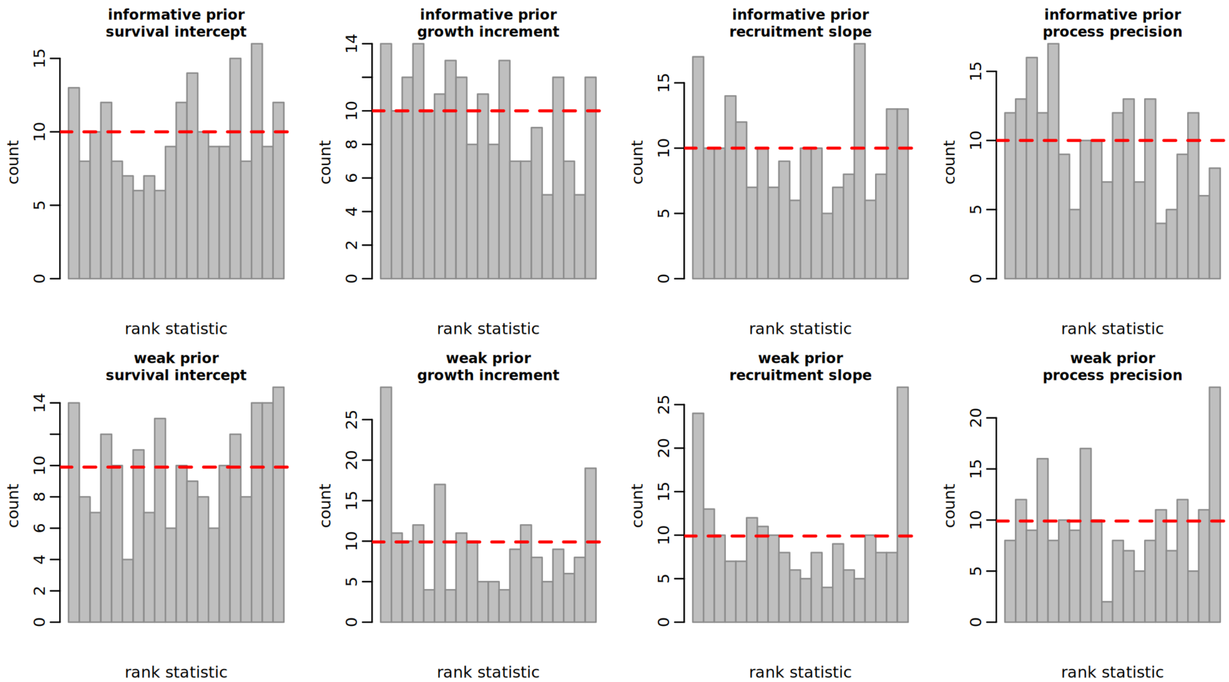

Figure 4: Simulation-based calibration rank histograms (dashed line: uniform expectation). Top: informative prior, ranks approximately uniform. Bottom: diffuse prior, U-shaped ranks indicating overconfident intervals, worst for the recruitment slope.

5.5 Observation-model robustness

A constant, size-independent detection probability is absorbed by the linear projection and does not bias the profile-only fit. Size-dependent detection does. We simulated three regimes—constant detection, unknown time-varying effort, and size-dependent detection—and fit the standard recovery model (Figure 5). Under constant detection, survival-at-80 coverage was 0.90–0.95; under time-varying effort it fell to about 0.65; under size-dependent detection it collapsed to zero, because a size-dependent change in detectability is absorbed as demographic change. This is the dominant real-world threat to profile-only inference and motivates the application caveats below.

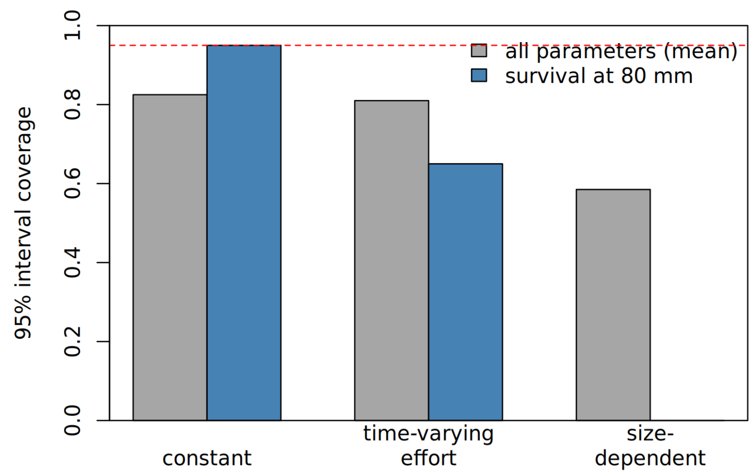

Figure 5: 95% interval coverage (Gamma–Poisson recovery) under three detection regimes. Size-dependent detection collapses survival coverage; the dashed line is the nominal 0.95 level.

5.6 Auxiliary data and transportability

Because recruitment intensity is the least identified rate, we quantified how much external individual-level fecundity information resolves it, expressed as an explicit auxiliary study of n_e fish analyzed by Poisson regression whose (approximate) posterior becomes the prior for (α_r, β_r). Correctly transportable external data improved recruitment recovery monotonically in n_e; 800 external fish cut recruitment-curve RMSE by about 70% and raised recruitment-at-80 coverage from 0.53 to 0.87. A precise but *biased* external study produced low point-estimate error yet zero coverage; robust (mixture) borrowing mitigated but did not eliminate the damage, and could reject a non-transportable study only when the conflict altered profile dynamics along a likelihood-informed direction (Figure 6). No single amount of external-study bias is universally safe.

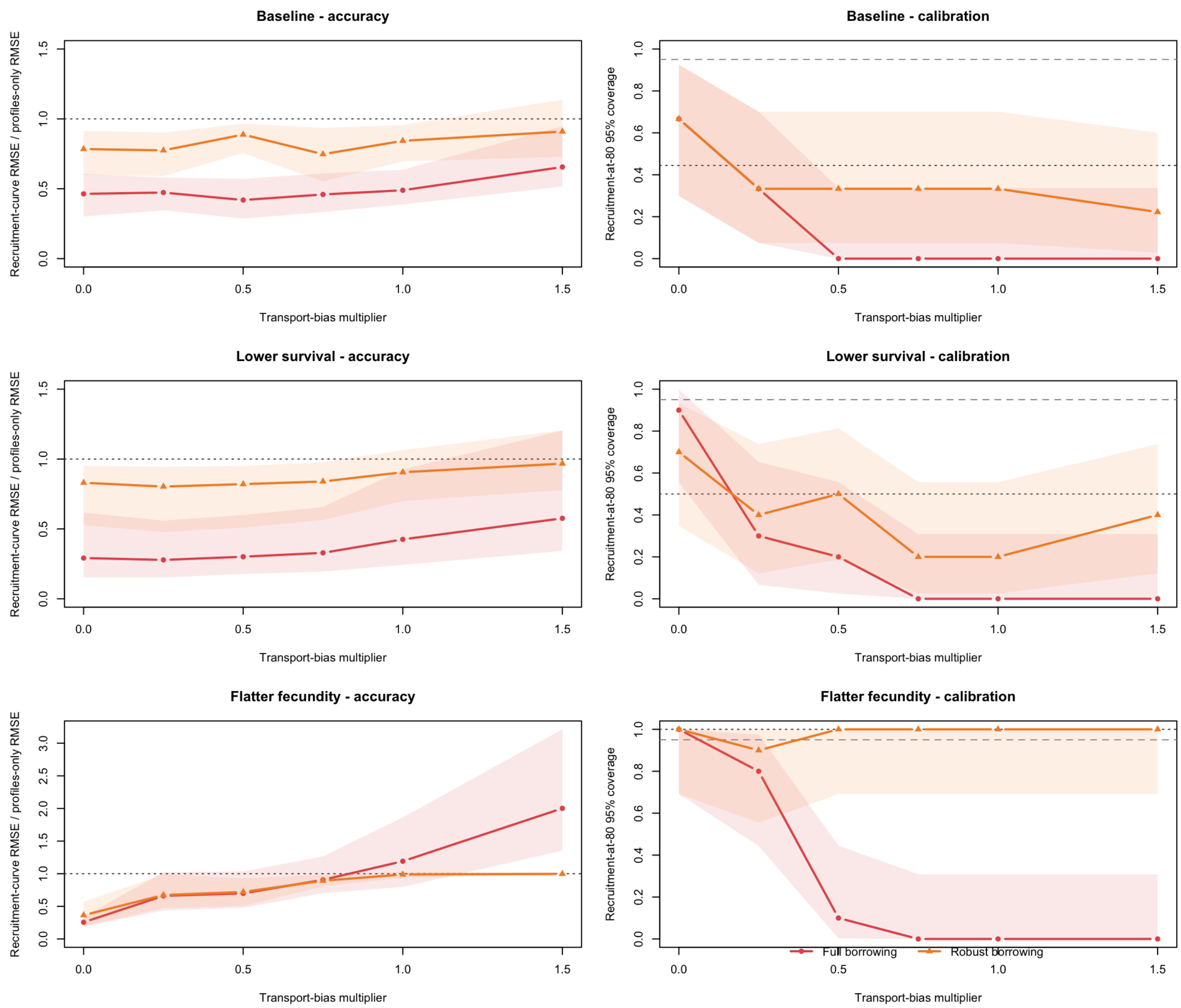

Figure 6: Accuracy and calibration as external-study bias increases (pooled across biological truths; 30 replicates per cell). Full borrowing improves point error but loses calibration at the smallest tested bias; robust borrowing degrades more gracefully. Dotted lines: profiles-only performance; dashed: nominal coverage.

5.7 Paired-data benchmark

Finally, on 200 datasets each carrying both profiles and capture–recapture histories, we compared profile-only, capture–recapture-only, integrated, and complete-history oracle analyses (Figure 7). Capture–recapture efficiently identified survival and growth but carried essentially no information about recruitment (its apparently perfect recruitment coverage reflected prior-width intervals, not recovery). Integration improved point accuracy for survival and growth but narrowed intervals below nominal, as residual discrepancy between observed and deterministically projected profiles was absorbed by other parameters—a caution we return to in the application.

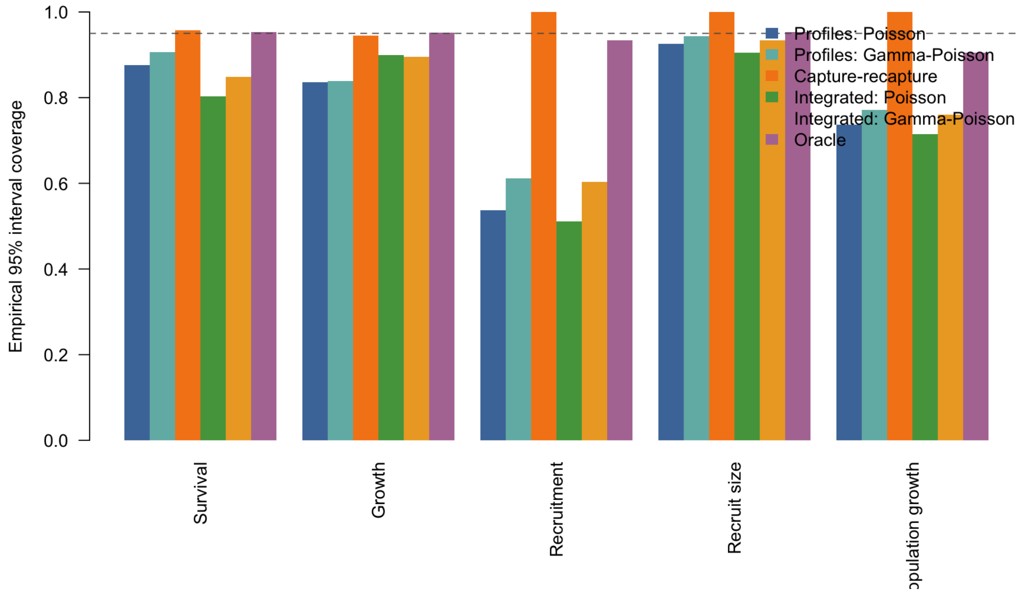

Figure 7: Empirical interval coverage in the paired-data benchmark. Perfect capture–recapture coverage for recruitment and population growth reflects prior-dominated wide intervals, not data-based identification.

6 Application: West Brook Brook Trout

6.1 Data and study system

We applied the framework to a long-term passive-integrated-transponder (PIT) tagging study of brook trout (*Salvelinus fontinalis*) in the West Brook, Massachusetts, USA (Letcher et al., 2015), obtained from the U.S. Geological Survey (U.S. Geological Survey, 2024). We treat the whole stream network (main stem and three tributaries) as one population, because a non-trivial fraction of fish move between reaches each year and a main-stem-only boundary would count emigrants as deaths. The profile stream is the standardized seasonal size cross-section over 2000–2019 (20 annual occasions, 19 transitions), restricted to sizes ≥ 70 mm where electrofishing detection is approximately size-uniform; recruitment is modeled as entry into this observable class. The size profiles are close to stationary across the two decades (Figure 8).

▶ **[COAUTHOR INPUT NEEDED – biology/ecology]:** *Expand this into a proper study-system description: stream characteristics, sampling design and electrofishing protocol, age/size structure of brook trout in this system, the biological meaning of "recruitment to the ≥ 70 mm class" here, and why the population may be near a stable size distribution. Add references to the field program.*

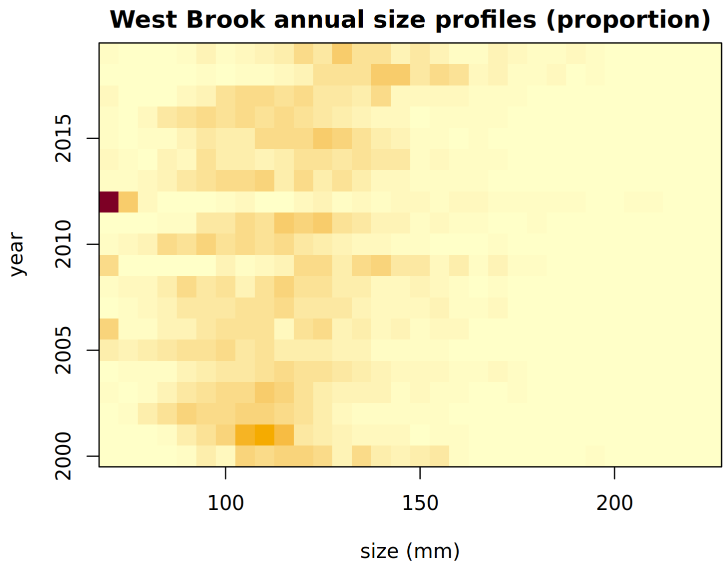

Figure 8: Observed annual size profiles (proportion by 5 mm bin) of West Brook brook trout, 2000–2019. The distribution is close to stationary across years.

6.2 Profile-only, mark–recapture, and integrated analyses

We fit the profile-only inverse IPM, a size-structured mark–recapture model for survival and growth (with detection estimated at $\hat{p} = 0.97$ from a constant-rate Cormack–Jolly–Seber model over the full histories, 20,077 recapture events), and an integrated model combining both likelihoods. Growth estimated from recaptures is detection-free and serves as the cleanest cross-check.

The profile-only fit is internally plausible—moderate survival, positive growth, $\lambda \approx 1$, a positive-definite posterior—yet it is contradicted by the individual data on both quantities the recaptures can check (Table 2, Figure 9). Profile-only inference *underestimates the growth increment* (12.6 vs. 32.3 $\mathrm{mm\,yr^{-1}}$), the signature of a near-stationary size distribution that hides how fast individuals move through it, and *misses the size gradient in apparent survival* (roughly flat near 0.44, versus a clear decline from 0.31 to 0.12 with size in the recapture and integrated analyses, reflecting greater emigration of larger fish). Recruitment intensity remains unidentified by profiles (level/slope posterior correlation -0.995), as in the simulations. The integrated analysis adopts the recapture survival and growth and tightens the population growth rate to $\lambda = 1.016\,[1.00, 1.05]$.

Table 2: West Brook brook trout: posterior median [95% interval] by analysis. Mark–recapture survival is *apparent* survival, which folds in size-dependent emigration from the network.

Quantity	Profile-only	Mark–recapture	Integrated
Survival @ 80 mm	0.43 [0.29, 0.60]	0.31 [0.31, 0.32]	0.31 [0.31, 0.32]
Survival @ 110 mm	0.45 [0.35, 0.55]	0.20 [0.20, 0.21]	0.20 [0.19, 0.21]
Survival @ 140 mm	0.46 [0.34, 0.58]	0.12 [0.12, 0.13]	0.12 [0.11, 0.12]
Growth incr. @ 80 mm	12.6 [9.3, 15.6]	32.3 [31.7, 32.8]	32.1 [31.5, 32.7]
λ	0.99 [0.97, 1.11]	—	1.016 [1.00, 1.05]

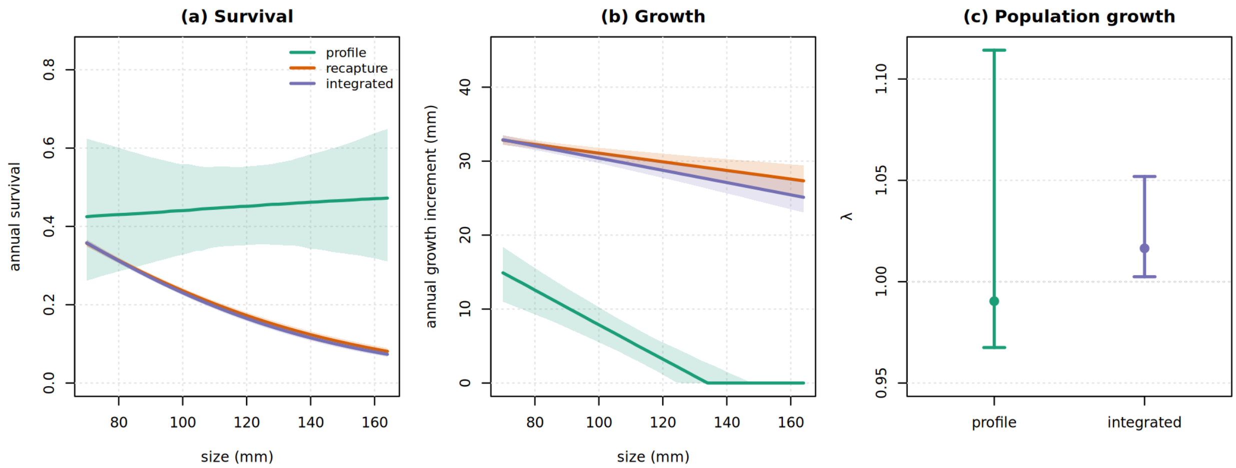

Figure 9: West Brook brook trout: (a) survival, (b) growth increment, and (c) population growth λ under profile-only, mark–recapture, and integrated analyses (medians with 95% bands/intervals). The profile-only fit underestimates growth and misses the survival size gradient that the individual data reveal.

▶ **[COAUTHOR INPUT NEEDED – biology/ecology]:** *Biological interpretation of these contrasts: is the $\sim$32 mm yr^{-1} recapture growth consistent with known brook trout growth in this system? Is the declining apparent survival with size attributable to emigration of larger fish, size-selective mortality, or both? What would the management or ecological reading of the profile-only versus integrated λ be? Please also confirm the framing of the size-dependent-detection caveat for electrofishing.*

6.3 Profile-only uncertainty is multimodal and concentrates on an iso-λ manifold

Replacing the Laplace approximation with full MCMC (four adaptive-Metropolis chains) exposes the non-identifiability directly. The chains do *not* converge on the vital rates (split $\widehat{R}$ up to 6.6); this is not a sampler failure but a property of the target. The chains settle in biologically distinct regimes—survival at 80 mm from 0.005 to 0.12, growth increment from 3 to 14 mm, recruitment over more than an order of magnitude—that all reproduce the near-stationary profiles of Figure 8. Yet across the pooled posterior the dominant eigenvalue varies by only 2.5% (coefficient of variation), versus 222% for survival at 80 mm and 131% for recruitment intensity (Figure 10): a direct, real-data instance of Proposition 2. The Laplace mode (survival ≈ 0.43) does not lie where the posterior mass concentrates, so its narrow intervals for the confounded directions are not trustworthy. We therefore report λ as the profile-identified functional and treat profile-only vital-rate point estimates as non-identified absent individual data; the integrated analysis both resolves the rates and shifts λ to 1.016 [1.00, 1.05].

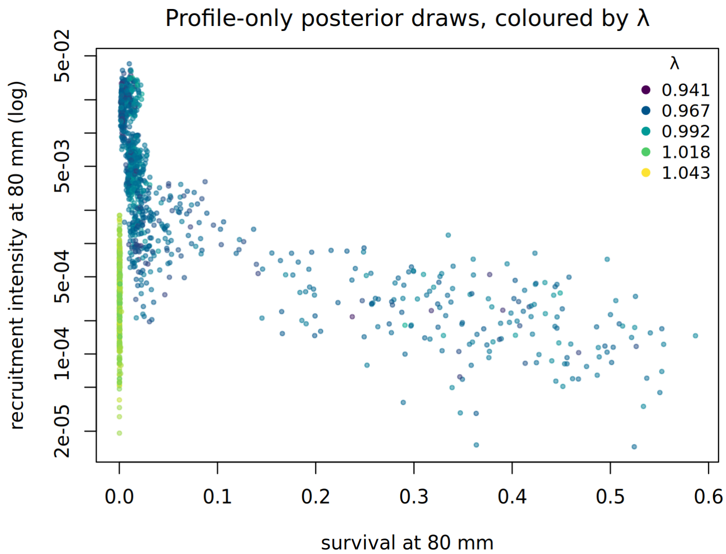

Figure 10: Profile-only posterior draws (West Brook brook trout) in the survival–recruitment plane, coloured by the implied dominant eigenvalue λ. Biologically very different vital-rate combinations span the plane, yet λ varies little (CV 2.5% vs. 222% for survival): profiles identify the asymptotic functional, not the rates that generate it.

This non-identification is invisible to local diagnostics. The observed Fisher information at the Laplace mode spans seven orders of magnitude, and nine of the eleven information eigen-directions contract by more than half from the prior (Figure 11); the single flat direction there is the recruit-size location, unconstrained only because recruitment into the window is near zero at that mode. A Hessian- or Laplace-based identifiability check would therefore declare the model essentially identified, even though the global posterior is the multimodal iso-λ manifold of Figure 10. Local curvature certifies practical identifiability *at a point*; establishing global identifiability requires manifold-revealing MCMC, profile likelihoods, or data cloning.

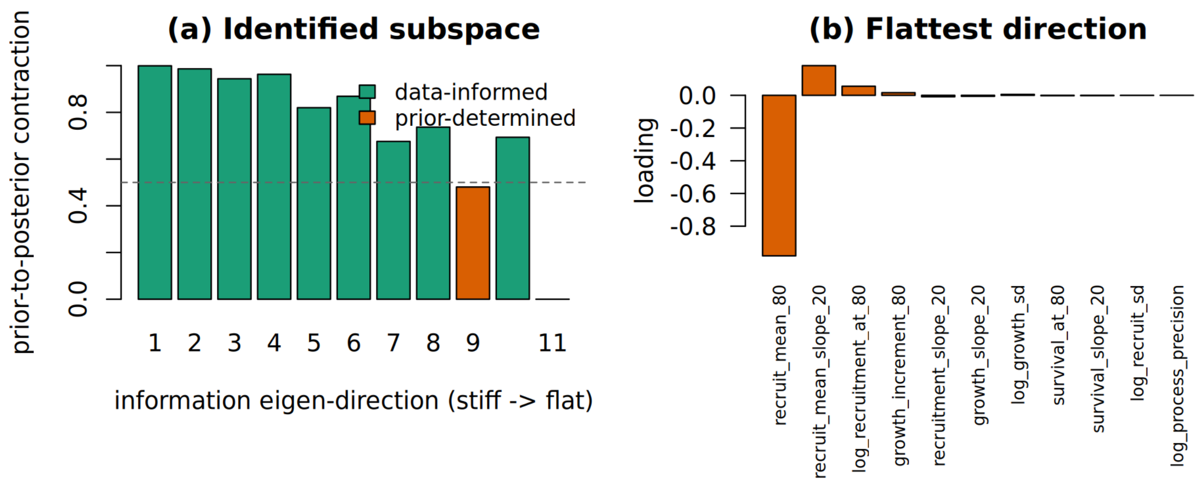

Figure 11: (a) Prior-to-posterior contraction along the observed-information eigen-directions at the West Brook profile-only mode; nine of eleven exceed 0.5, suggesting local identifiability. (b) The flattest direction loads almost entirely on recruit-size location. Local curvature does not reveal the global non-identification of Figure 10.

7 Discussion

Repeated size profiles are genuinely informative about some vital rates—survival, the growth distribution, and recruit size—and almost uninformative about others, chiefly recruitment intensity and, through it, the asymptotic growth rate. The confounding is structural (a ridge that trades recruitment level against its size gradient), it is sharpened by short series and near-stationary distributions, and it is easily masked by an informative prior or by unmodeled size-dependent detection. The brook trout application makes the practical stakes concrete: a profile-only fit that passes every internal check still misstates growth by a factor of two and inverts the survival size pattern, errors visible only against individual data.

For practice we recommend: declaring the estimand before fitting; reporting prior-to-posterior contraction and the recruitment level/slope correlation as routine identifiability diagnostics; modeling, or at minimum bounding, size- dependent detection; and expressing external information as an explicit, transportability-checkable study design. Where individual data exist, an integrated analysis is worth far more than its marginal cost, but its intervals must be interpreted cautiously when the profile and individual components disagree.

Limitations include reliance on a parametric vital-rate basis (flexible spline bases are a direct extension), a closed-population deterministic projection likelihood (the finite-population particle-filtered variant is developed separately), and a single applied system. The framework and diagnostics, however, are general to size- and stage-structured inverse problems.

▶ **[COAUTHOR INPUT NEEDED – biology/ecology]:** *Two to three sentences on the ecological/management implications and on how these recommendations would change analyses in your systems, plus any domain-specific limitations we should acknowledge.*

Data and Code Availability

The West Brook data are public domain (CC0) and available from the U.S. Geological Survey (U.S. Geological Survey, 2024); a fetch script is provided. All simulation code, analysis scripts, and figure-generation code are in the project repository.

Acknowledgments

▶ **[COAUTHOR INPUT NEEDED – biology/ecology]:** *Funding sources, field crews, and data-collection acknowledgments for the West Brook program.*

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